3. Output parameterization and fixed architecture

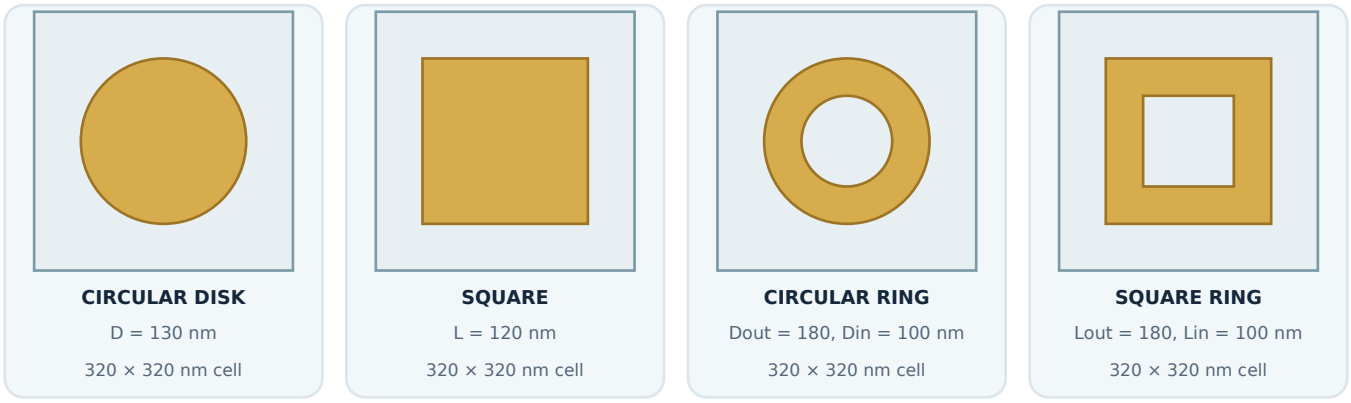


Figure 2. Four candidate geometries, all with unrestricted size within the 320 nm periodic cell. The same mathematical decoder covers disk/ring and square/square-ring families.

Use two normalized coordinate outputs and a deterministic full-domain geometry mapping. Unlike a sigmoid-only decoder, a hard-sigmoid can represent the exact 0 and 1 endpoints, including zero-size and 320 nm contact configurations.

$$d_{\text{out}} = 320 z_o, \qquad d_{\text{in}} = d_{\text{out}} z_i$$

$$z_o = \text{hardsigmoid}(u_o), \quad z_i = \text{hardsigmoid}(u_i)$$

For circles, d is diameter; for squares, d is side length. Solid shapes use d_out = 320 z_o and fix d_in = 0. Rings use both outputs. No geometric rounding or minimum wall/gap is applied. For d_out = 0 or d_in = d_out, the CAD builder must represent zero Au volume without creating a degenerate solid. For d_in = 0, a ring is physically the corresponding solid. The network may return endpoints exactly.

Architecture, selected once

Component	Fixed configuration
Input vector	(a − 450)/1150, (b − 450)/1150, X (3 normalized scalars)
Shared trunk	Linear(3,256) → SiLU → 3 residual blocks (width 256, LayerNorm, SiLU)
Category head	Linear(256,8), softmax over 4 families × 2 environments
Dimensions head	Linear(256,2) → hard-sigmoid → full-domain mapping
Single model	One shared trunk and two heads, trained jointly
Precision	FP32 training; no quantization requirement